

AMIT MAHAJAN

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PROFILE SUMMARY

M.Sc. Computer Applications graduate with hands-on experience building AI-powered applications using Python, LangChain, LangGraph, Retrieval-Augmented Generation (RAG), and LLMs. Experienced in developing multi-agent workflows, integrating external APIs, and building end-to-end AI applications with Django and Streamlit. Strong foundation in Machine Learning, prompt engineering, and scalable Python development, with a passion for building production-ready AI systems.

SKILLS

Languages & Database: Python, HTML5, CSS3, JavaScript, MySQL .

Libraries / Frameworks: Django, FastAPI, Streamlit, LangChain, LangGraph, Scikit-learn, TensorFlow (Basics), PyTorch (Basics).

ML : Supervised & Unsupervised Learning, Classification, Regression, Clustering, Model Training, Model Evaluation, Data Preprocessing

Generative AI & LLM : LangChain, LangGraph, Large Language Models (LLMs), AI Agents, Retrieval-Augmented Generation (RAG), Prompt Engineering, Multi-Agent Systems, Embeddings, Semantic Search, Vector Databases (FAISS/ChromaDB - Basics).

Tools: Git, GitHub, REST APIs, Jupyter Notebook, VS Code

EXPERIENCE

1. AI & ML Intern (Remote): Elevate Labs– Bengaluru. January 2026 – April 2026

Key Responsibilities:

- o Performed **data preprocessing and feature engineering** on real-world datasets.
- o Developed production-ready Machine Learning models using Python and Scikit-learn following end-to-end ML workflows.
- o Applied **classification and regression algorithms** for predictive analytics.
- o Conducted **model evaluation** using accuracy, precision, recall, and F1-score.
- o Utilized **NumPy, Pandas, and Matplotlib** for data analysis and visualization.
- o Collaborated using **Git** and **GitHub** while maintaining clean, modular, and reusable Python code.

Tools & Technologies: Python, Machine Learning, Data Analysis, Model Development, Jupyter Notebook.

PROJECTS

1 Multi-Disease Prediction System | Django, Python, Scikit-learn, Machine Learning

- o Developed an AI-powered web application for predicting **Diabetes, Heart Disease, Kidney Disease, and Parkinson's Disease** using Machine Learning models integrated into a **Django** application.
- o Performed **data preprocessing, feature selection, and model training** on healthcare datasets using **Scikit-learn** and **Python**.
- o Implemented classification algorithms including **Logistic Regression, Random Forest, and Support Vector Machine (SVM)** and evaluated model performance using **accuracy metrics**.
- o Serialized trained models using **Pickle (.pkl)** files and deployed them within the **Django** application for **real-time disease prediction**.
- o Developed secure **user authentication, prediction history management, and downloadable PDF** report generation features.
- o Designed a responsive user interface using **Django Templates**, HTML, CSS, and Bootstrap following a modular and scalable architecture.

2. AskBuddy | AI-Powered QnA Chatbot

View Project : <https://askbuddychatbot.streamlit.app/>

- o Developed an AI-powered multi-agent Question Answering system using **LangChain** and **LangGraph** with **contextual memory** and **Retrieval-Augmented Generation (RAG)**.
- o Integrated **Google Serper Search API** as an external retrieval tool enabling real-time knowledge retrieval within an agent workflow.
- o Designed a LangGraph multi-agent workflow with **MemorySaver** to maintain contextual memory and coordinate agent execution across multiple conversation turns.
- o Enabled **real-time response streaming** and **session management** to improve chatbot responsiveness and user experience.
- o Developed an end-to-end AI application with **Streamlit** supporting agent execution, **conversational memory, response streaming, and external API integration**.

Tech Stack: Python, LangChain, LangGraph, Streamlit, Groq LLM, Google Serper API, Generative AI, Prompt Engineering

EDUCATION

Dr D Y Patils ACS College , Pimpri | M.Sc. Computer Application | SPPU University | Expected 2026

Dr. D.Y.Patil SCS College , Akurdi | BSC in Computer Science | SPPU University | 2024 | CGPA – 8.67

CERTIFICATIONS

Python – Exponent IT Training Institute | Data Science and Data Analysis – Exponent IT Training Institute